Answer Key – CYBER WORLD AND SOCIAL REALITIES

(Answer any 5 – 2 marks each)

1. What is Virtual Reality (VR), and how does it create immersive experiences?

Virtual Reality (VR) is a computer-generated three-dimensional environment that allows users to interact with a simulated world using devices such as VR headsets and sensors. It creates immersive experiences by stimulating vision, sound, and movement, making users feel physically present in a virtual environment.

2. What factors influence how people curate their online identities?

Online identities are influenced by factors such as social expectations, peer influence, personal interests, cultural background, privacy concerns, and the desire for social acceptance. Social media platforms also shape how individuals present themselves online.

3. How does technology impact the development of moral values in individuals and societies?

Technology influences moral values by shaping communication, behaviour, and social interaction. It can promote awareness, education, and global understanding, but excessive dependence on technology may reduce face-to-face interaction and encourage unethical practices such as cyberbullying, plagiarism, and online fraud.

4. Discuss the importance of digital identity in today's society.

Digital identity refers to the online presence and personal information associated with an individual. It is important because it enables communication, online transactions, education, employment opportunities, and access to digital services. A secure digital identity also helps build trust and social participation.

5. What are social networks, and how do they create online communities?

Social networks are digital platforms that connect people through communication and sharing of information. They create online communities by allowing users with common interests, goals, or identities to interact, collaborate, and exchange ideas regardless of geographical boundaries.

6. What are the ethical concerns surrounding the use of AI in society?

Major ethical concerns related to AI include privacy violations, data misuse, job displacement, algorithmic bias, lack of accountability, surveillance, and the possibility of misuse in warfare or manipulation. These concerns highlight the need for ethical regulation of AI technologies.

Part B – Short Essay Questions

(Answer any 2 – 5 marks each)

7. Discuss the Negative Impact of AI

Artificial Intelligence (AI) has many advantages, but it also creates several negative impacts on society.

Job Displacement – Automation may replace human workers in industries, leading to unemployment.
Privacy Concerns – AI systems collect large amounts of personal data, increasing the risk of surveillance and misuse.
Bias and Discrimination – AI algorithms may reflect social biases and produce unfair decisions.
Cybercrime and Misuse – AI can be used for hacking, deepfakes, and spreading misinformation.
Dependence on Technology – Excessive reliance on AI may reduce human creativity and decision-making abilities.
Ethical Issues – Lack of accountability in AI decisions raises moral and legal concerns.

Therefore, responsible use and regulation of AI are essential for protecting society.

8. List and Explain Some Examples of Respectful and Positive Behaviour Online

Positive online behaviour promotes healthy digital interaction and cyber ethics.

Examples:
Using Polite Language
Communicating respectfully without abusive or offensive words.
Respecting Privacy
Not sharing others' personal information or images without permission.
Avoiding Cyberbullying
Treating others kindly and not harassing or humiliating people online.
Sharing Accurate Information
Verifying facts before forwarding messages or news.

Acknowledging Others' Opinions
Respecting different viewpoints and engaging in constructive discussions.
Practicing Digital Responsibility
Following ethical rules and reporting harmful online activities.

Such behaviours help create safe and inclusive online communities.

9. What is the Concept of a Network Society, and What are its Defining Characteristics?

The concept of a network society was developed by Manuel Castells. A network society is a society where social, economic, and cultural activities are organized through digital communication networks.

Characteristics:
Global Connectivity – People are connected across the world through the internet.
Information-Based Economy – Knowledge and information become key economic resources.
Digital Communication – Online interaction replaces many traditional forms of communication.
Flexibility and Decentralization – Networks allow flexible and rapid exchange of information.
Virtual Communities – Social relationships increasingly occur in digital spaces.

The network society has transformed work, education, politics, and social relationships.

10. How does Identity Theft Occur in the Digital World, and What can Individuals do to Safeguard their Personal Information?

Identity theft occurs when criminals steal personal information such as passwords, bank details, Aadhaar numbers, or social media credentials for fraudulent purposes.

Common Methods:
Phishing emails and fake websites
Hacking and malware attacks
Data breaches
Social engineering techniques
Safeguarding Measures:
Use strong and unique passwords.
Enable two-factor authentication.
Avoid sharing sensitive information online.
Update software and antivirus systems regularly.
Be cautious while opening suspicious emails or links.
Use secure websites for online transactions.

Digital awareness and cyber security practices help protect individuals from identity theft.

Part C – Essay Type Questions

(Answer any 1 – 15 marks)

11. Examine the Potential Benefits and Challenges of Implementing AR and VR Technologies in Various Sectors of Society

Augmented Reality (AR) and Virtual Reality (VR) are advanced technologies that transform how people interact with digital environments. AR overlays digital information onto the real world, while VR creates fully immersive virtual environments.

Benefits of AR and VR
1. Education

AR and VR improve learning through interactive and immersive experiences. Students can conduct virtual experiments, explore historical places, and understand complex concepts more effectively.

2. Healthcare

VR is used in medical training, surgery simulation, and therapy for mental health conditions. AR assists doctors during surgeries by displaying digital information in real time.

3. Entertainment and Gaming

These technologies provide immersive gaming and entertainment experiences, increasing user engagement.

4. Business and Industry

Companies use AR and VR for employee training, product design, virtual meetings, and marketing.

5. Tourism and Real Estate

Virtual tours help people explore tourist destinations and properties remotely.

6. Military and Defence

VR-based simulations are used for training soldiers in safe environments.

Challenges of AR and VR

1. High Cost

The equipment and software required for AR and VR are expensive.

2. Privacy and Security Risks

These technologies collect personal and behavioural data, raising privacy concerns.

3. Health Issues

Excessive use may cause eye strain, dizziness, headaches, and social isolation.

4. Digital Divide

Not everyone has equal access to advanced technologies, increasing inequality.

5. Ethical Concerns

Virtual environments may encourage addiction, misinformation, or harmful content.

6. Technical Limitations

Limited infrastructure, internet connectivity, and technical knowledge can restrict implementation.

12. Discuss the Features of Various Social Media Platforms and its Impact on Social Life

Social media platforms are digital communication tools that enable people to create, share, and exchange information globally.

Features of Social Media Platforms

1. Facebook

Allows users to connect with friends, share posts, photos, videos, and join groups and communities.

2. Instagram

Focuses mainly on photo and video sharing with features such as stories, reels, and live streaming.

3. WhatsApp

Provides instant messaging, voice calls, video calls, and group communication.

4. Twitter/X

Used for sharing short messages, opinions, and news updates quickly.

5. YouTube

A video-sharing platform for education, entertainment, and information dissemination.

6. LinkedIn

A professional networking platform used for career development and employment opportunities.

Impact of Social Media on Social Life
Positive Impacts
Improved Communication – Connects people across geographical boundaries.
Access to Information – Provides instant news and educational content.
Social Awareness – Helps spread awareness about social issues and movements.
Business Opportunities – Supports digital marketing and entrepreneurship.
Community Building – Creates support groups and online communities.
Negative Impacts
Addiction and Time Wastage – Excessive use affects productivity and mental health.
Cyberbullying – Online harassment negatively affects individuals.
Privacy Risks – Personal data may be misused or leaked.
Fake News and Misinformation – Rapid spread of false information creates confusion.
Reduced Face-to-Face Interaction – Overdependence on social media weakens direct social relationships.